Applied AI

Lecture 1

Dr Artie Basukoski

Agenda

- Introduction
- Assessment
- Resources
- Lectures
- Recent Developments
- History, issues
- Next steps

Introduction

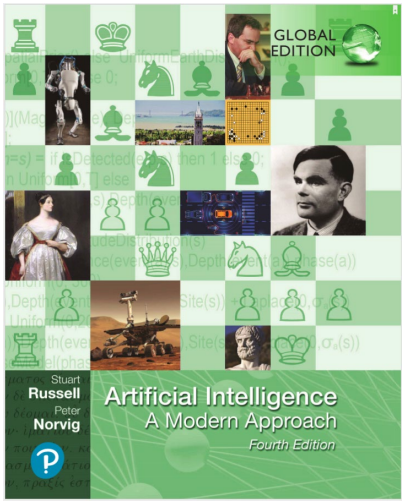
Aims: To give the student the background knowledge and practical skills to use and evaluate AI techniques over a range of problem domains.

- Understand the fundamental concepts, issues, and techniques of AI.
- Implement and use AI techniques across a broad range of AI sub-fields.
- Evaluate when and under what conditions it is appropriate to use each AI technique.
- Understand the historical background and evolution of AI techniques.
- Each week an essential technique will be demonstrated via a working implementation followed by a presentation of the theory and conditions needed to enable the student to set up and use the techniques themselves. Guest presentations by experts in a relevant topic will be arranged where possible.

Assessment

Assessment Name	Weight	Qual Mark	Due Date	
Coursework	50	30	Wed, 6th Jan 1pm	Research, implement, and evaluate an AI application.
In-Class Test (On Blackboard)	50	30	Week 12 In your seminar (must attend in person or seek MC)	Multiple choice and short answer questions on the blackboard. Covering all topics up to week 11. Duration 90 minutes.

Resources

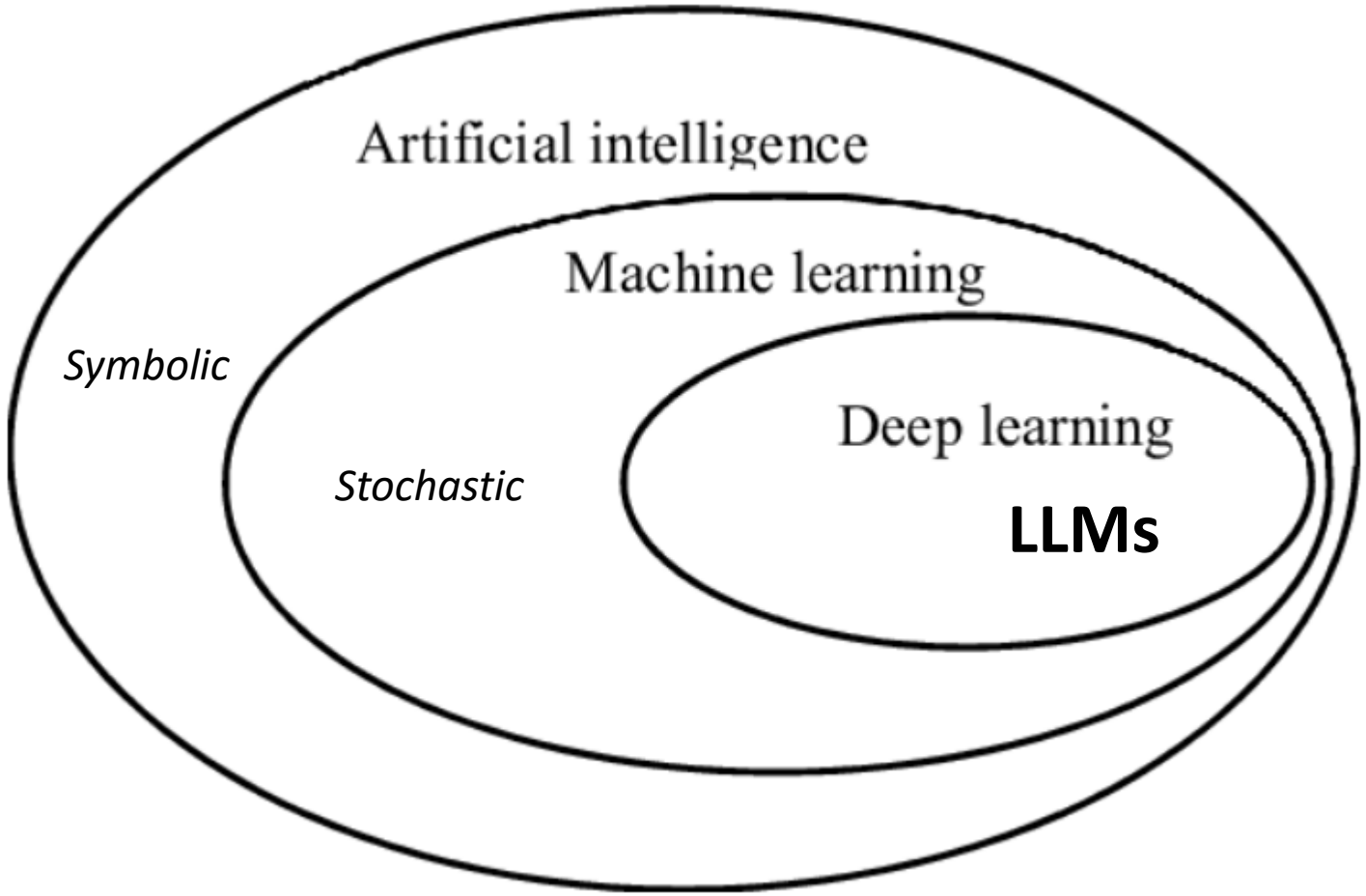


- Your main resource will be the book
  Artificial Intelligence: A Modern Approach, Russel and Norvig
  - Access via the **Reading List on Blackboard**
- Sample code in many languages is available on the book website:
- https://github.com/aimacode
- The web (obviously). There are many online courses and tutorials covering every aspect of artificial intelligence.
- arXiv.org approx. 120K freely available papers on Artificial Intelligence
- Additional resources will be added during the specific lectures.

Lecture schedule

1. Introduction and History of AI
2. Search
3. Adversarial Games
4. Agents and Environments
5. Knowledge representation and planning
6. -------Engagement week--------
7. Machine Learning
8. Neural Networks
9. Computer Vision/Deep Learning
10. LLMs
11. Reinforcement Learning
12. Revision session

Artificial Intelligence – broad overview



What is Artificial Intelligence?

- Intelligence: the ability to learn and perform suitable techniques to solve problems and achieve goals, appropriate to the context, in an uncertain, ever-varying world. (Prof. Chris Manning, Stanford)
- A term coined by Professor John McCarthy in 1955, was defined by him as "the science and engineering of making intelligent machines"
- Purposely open definition to encourage exploration. This has had positive and negative implications. See: Lighthill report (consider AGI redefinition)
- Tom Mitchell: "A computer program is said to learn from experience **E** with respect to some class of tasks **T** and performance measure **P**, if its performance at tasks in **T**, as measured by **P**, improves with experience **E**."
- See more definitions here: https://hai.stanford.edu/sites/default/files/2020-09/AI-Definitions-HAI.pdf

A (Short) History of AI

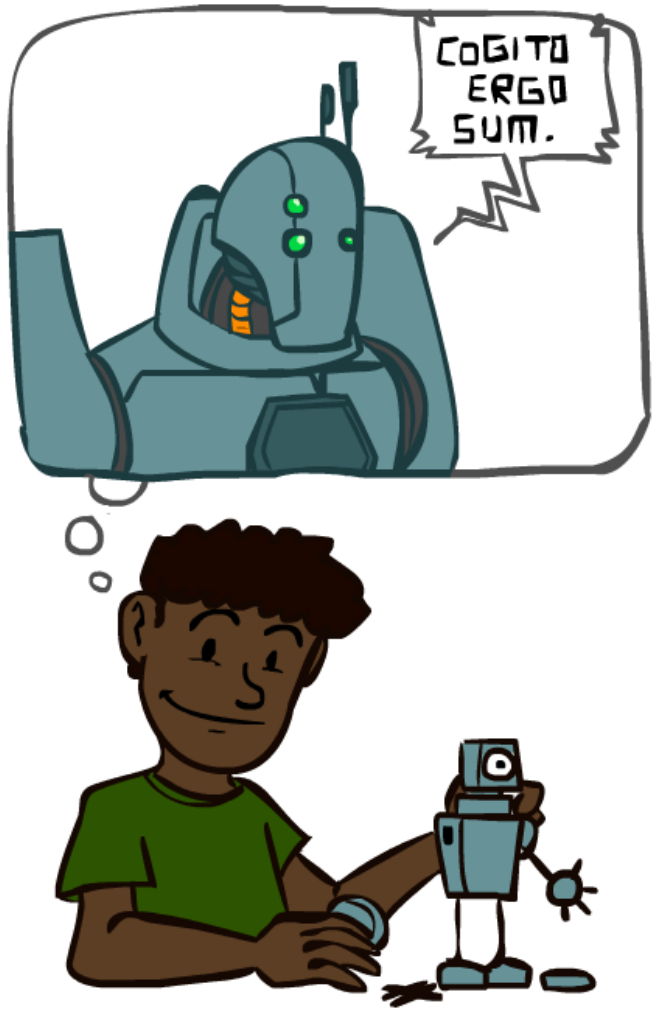
- 1940—1950: Early days
  - 1943: McCulloch & Pitts: Boolean circuit model of brain
  - 1950: Turing's “Computing Machinery and Intelligence”

- 1950—70: ***Excitement: Look, Ma, no hands!***
  - 1950s: Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
  - 1956: Dartmouth meeting: “Artificial Intelligence” adopted
  - 1965: Robinson's complete algorithm for logical reasoning

- 1970—90: Knowledge-based approaches
  - 1969—79: Early development of knowledge-based systems
  - 1980—88: Expert systems industry booms
  - 1988—93: Expert systems industry busts: “AI Winter”

- 1990—2012: Statistical approaches + subfield expertise
  - Resurgence of probability, focus on uncertainty
  - General increase in technical depth
  - Agents and learning systems… “AI Spring”?

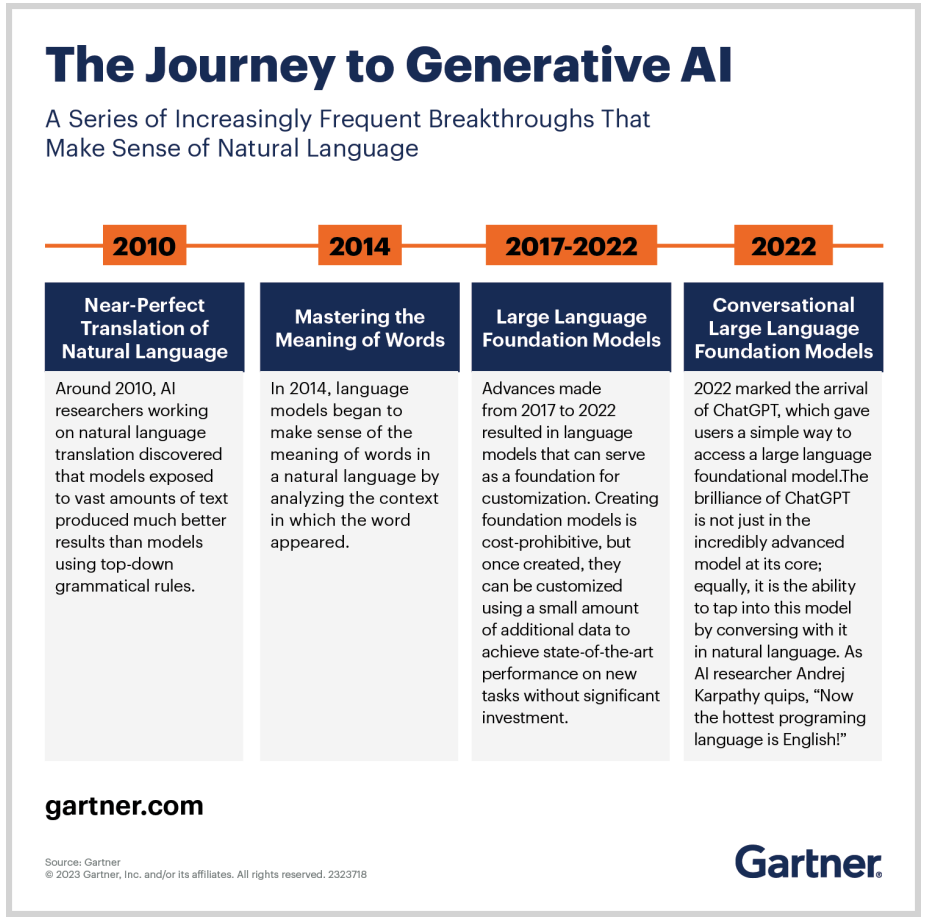
- 2012—: ***Excitement: Look, Ma, no hands!***
  - Big data, big compute, neural networks
  - Some re-unification of subfields
  - AI used in many industries



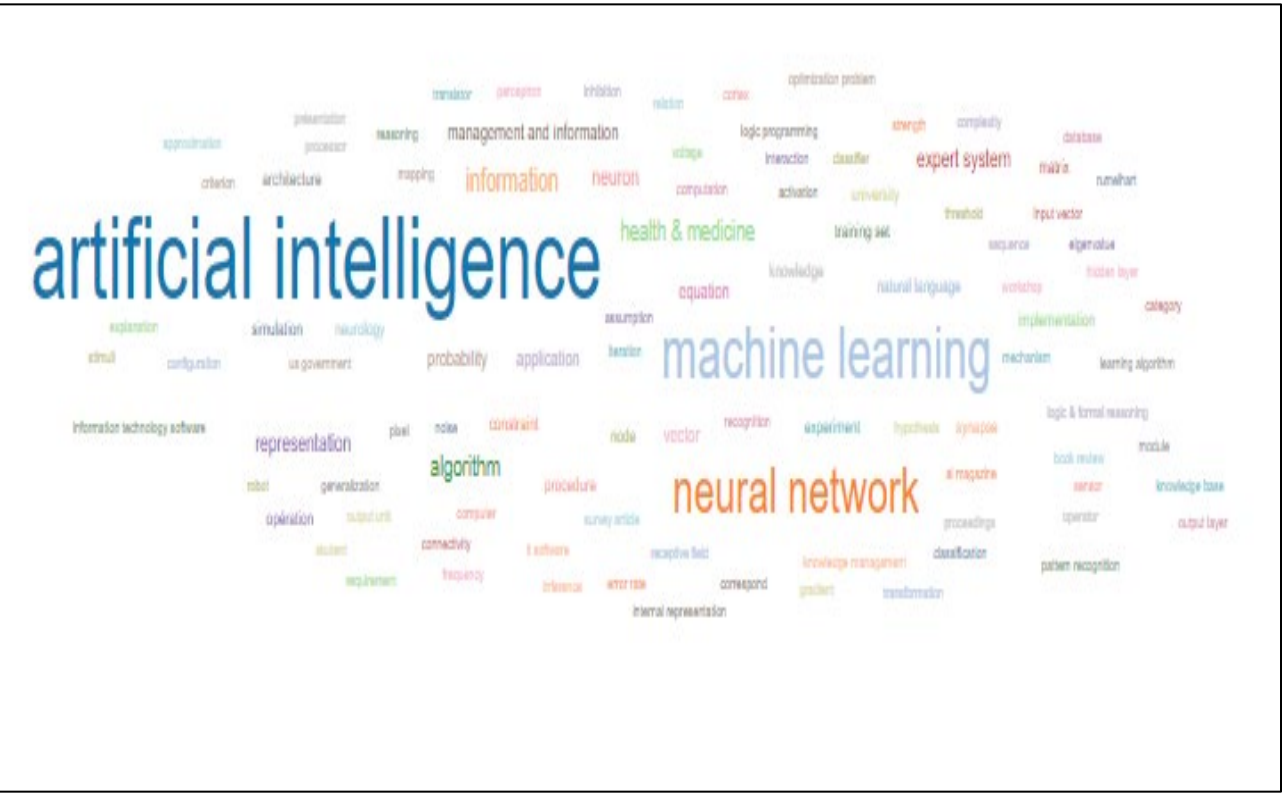
Source CS188 Lectures

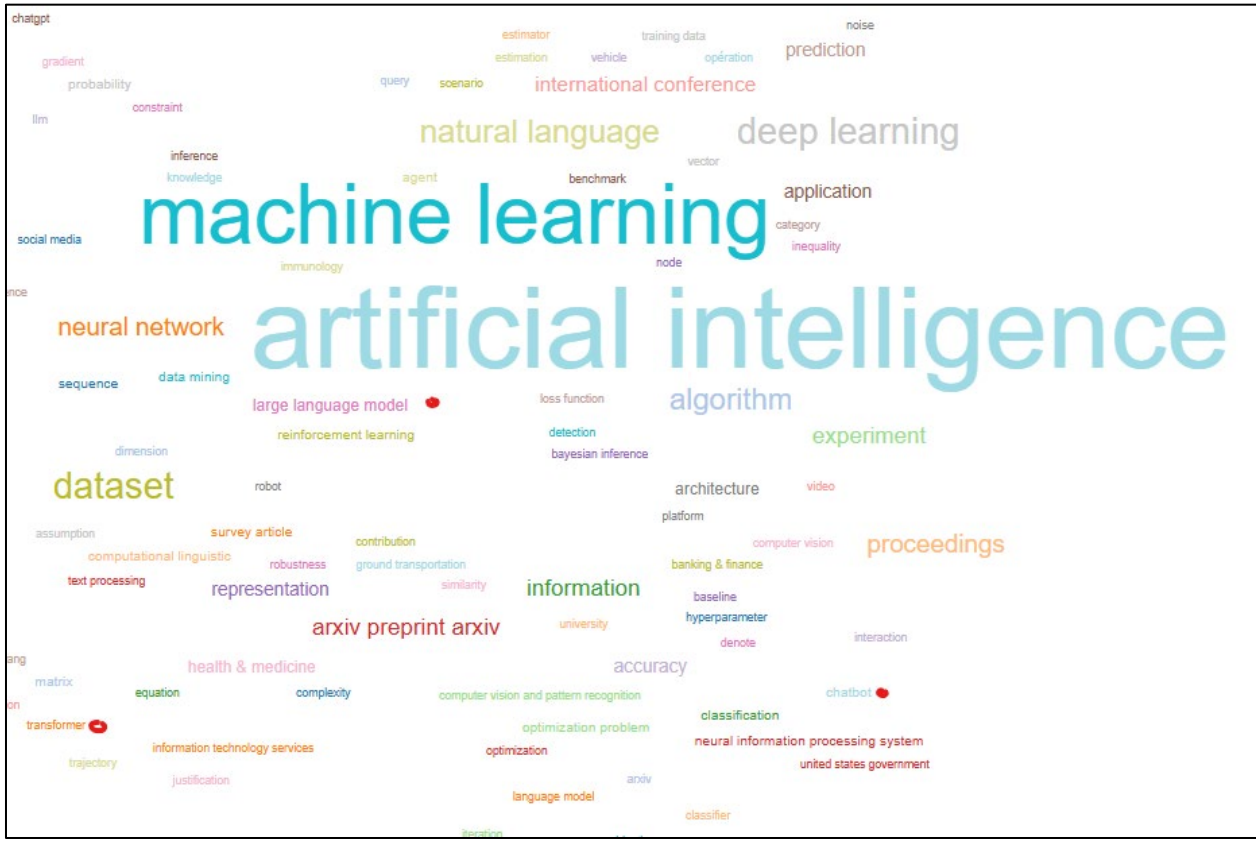
What is Artificial Intelligence - really

- **Problem Solving and Search**
  - Techniques: Uninformed and informed search, adversarial search, constraint satisfaction.
  - Applications: Pathfinding, game playing, optimization problems.
- **Knowledge Representation and Reasoning**
  - Techniques: Logic, ontologies, semantic networks, probabilistic reasoning.
  - Applications: Expert systems, natural language understanding, decision support.
- **Machine Learning**
  - Techniques: Supervised, unsupervised, reinforcement learning**, neural networks**.
  - Applications: Classification, regression, pattern recognition, predictive analytics.
- **Natural Language Processing (NLP)**
  - Techniques: Parsing, **machine translation**, **speech recognition**, sentiment analysis.
  - Applications: **Chatbots**, virtual assistants, language modeling.
- **Robotics and Perception**
  - Techniques: **Computer vision**, motion planning, sensor integration, control theory.
  - Applications: Autonomous vehicles, industrial robots, assistive technologies.
- **Planning and Decision Making**
  - Techniques: Classical and hierarchical planning, Markov decision processes, game theory.
  - Applications: Automated scheduling, logistics, strategic planning.
- **Uncertainty and Probabilistic Models**
  - Techniques: Bayesian networks, Markov models, decision theory.
  - Applications: Diagnostics, risk assessment, predictive maintenance.
- **AI Ethics and Safety**
  - Topics: Fairness, transparency, accountability, ethical implications of AI.
  - Focus: Responsible AI deployment, societal impact, regulatory considerations.
- **AI Philosophy and Future Directions**
  - Topics: Philosophical implications, strong vs. weak AI, future of AGI.
  - Focus: Challenges and opportunities in advancing AI towards general intelligence.



AI - 1986-1990 vs 2020-now





https://aitopics.org/search#

What are the main differences?
Explore other periods on your own.

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Which program was an early AI demonstration of logical reasoning?

A. ELIZA

B. Logic Theorist

C. General Problem Solver

D. Expert systems

Recent Developments Companies to watch

- OpenAI, Anthropic, huggingface
- DeepMind
- Tesla and Xai
- Microsoft
- Meta(Facebook)
- Amazon
- IBM



Recent Developments: Google Deep Mind

- In April 2023, Google announced that it is merging its two main AI research groups - Google Brain and DeepMind - into a single unit called Google DeepMind
- Pushing the frontier – see Genie 3



Recent Developments: LLMs, ChatGPT, Gemini, Llama, etc…

- OpenAI ChatGPT
- Google Gemini
- Perplexity
- Open Source
- Multi modal – see, hear, and interact!

- Controversial:Let's discuss
  - is it sentient?
  - what are the limitations?



Recent Developments: Natural Language to code

- Is software development just about code?
- Used to help developers become more productive and efficient, rather than to replace them entirely
- Code generators integrated with development environment. Eg, Github Copilot, cursor
- Intense competition

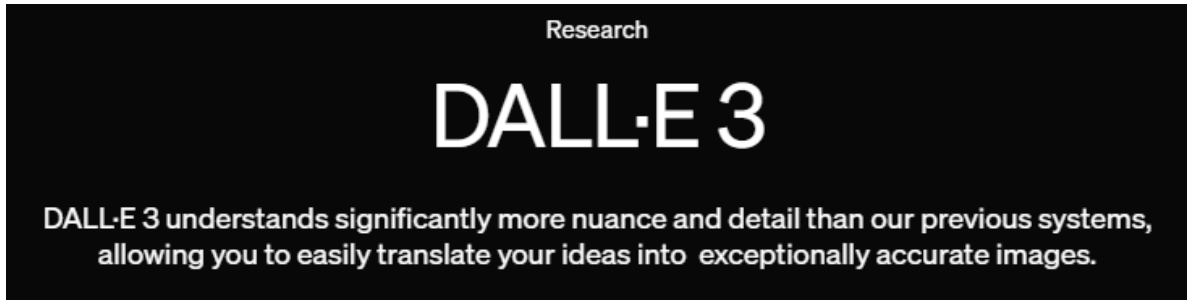
  Claude Code vs OpenAI Codex vs Copilot

- https://openai.com/codex/
- https://github.com/features/copilot
- https://claude.com/product/claude-code



Recent Developments: Natural Language to ART

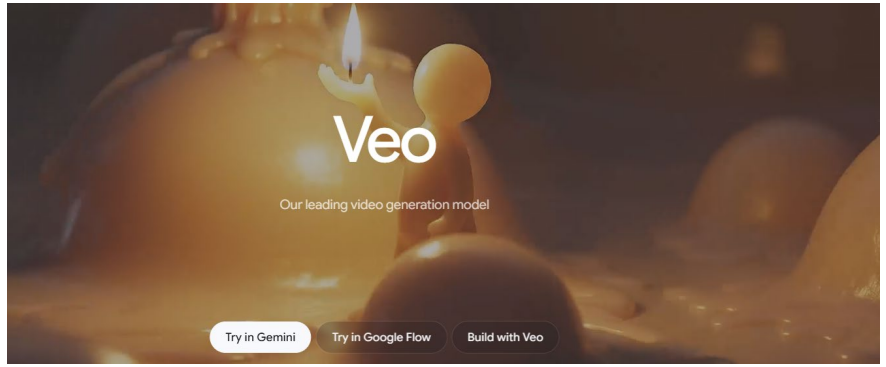
- Open AI
  - https://openai.com/index/dall-e-3/



- Stable Diffusion (open source)
  - https://stability.ai/news/
  - Try it: https://huggingface.co/spaces/stabilityai/stable-diffusion



- Google DeepMind (Veo, Omni, Lyria)
  - https://deepmind.google/models/veo/
  - Try it: https://flow.google.com/







Google - Nano Banana – Studio Ghibli style and Muppet

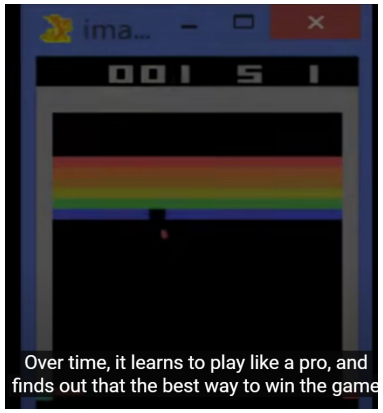
Recent Developments: Games:

- AlphaZero: Chess, Go
  - https://www.deepmind.com/blog/alphazero-shedding-new-light-on-chess-shogi-and-go

- 57 Atari games at superhuman level. https://www.youtube.com/watch?v=dJ4rWhpAGFI

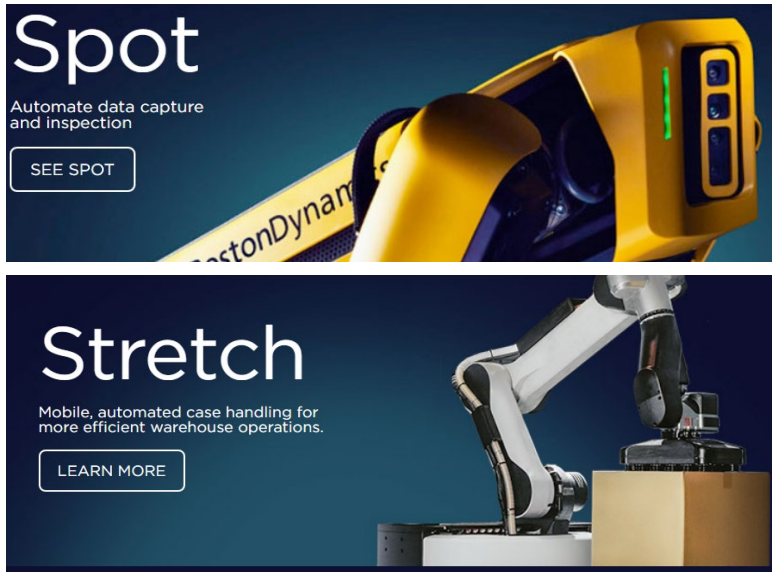
- Can beat top human players at the strategy game StarCraftII
  - https://www.nature.com/articles/d41586-019-03298-6



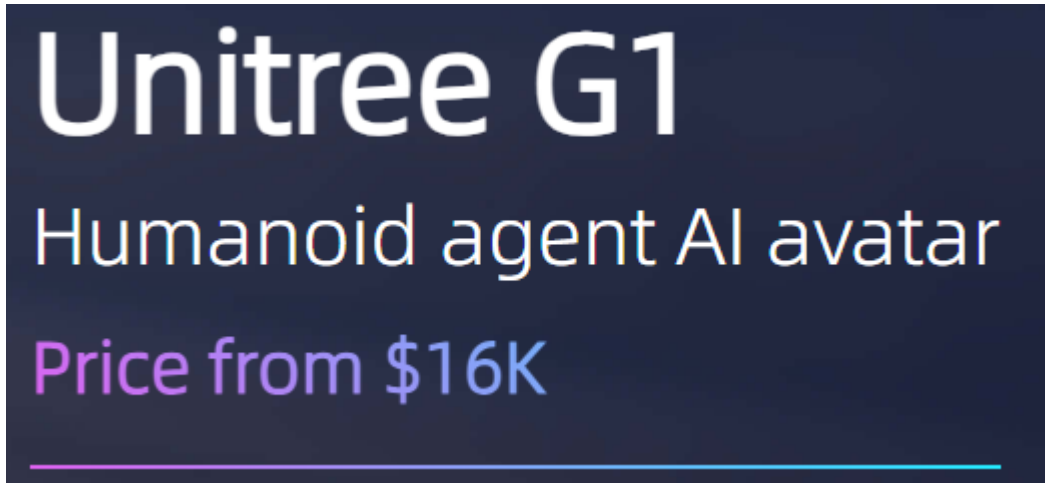




Recent Developments: Robotics



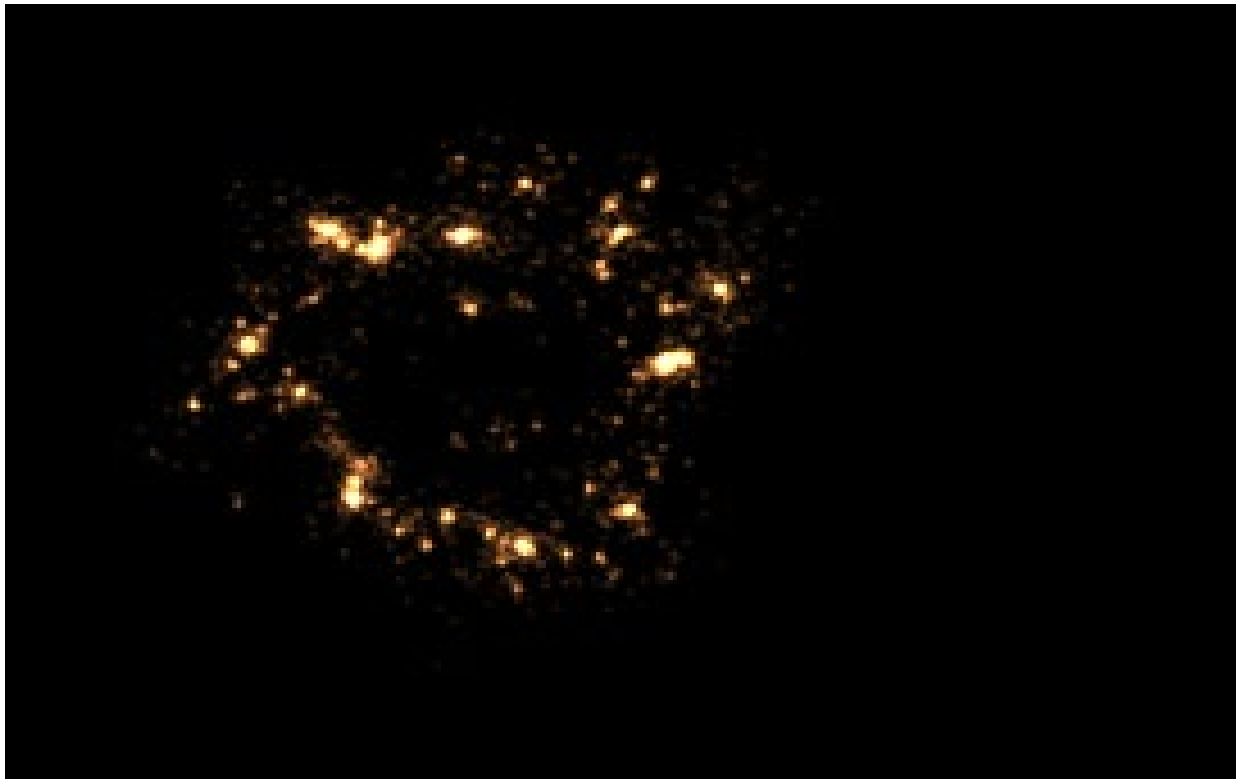
https://www.youtube.com/watch?v=-e1_QhJ1EhQ
https://www.youtube.com/watch?v=LMPxtcEgtds&t=65s



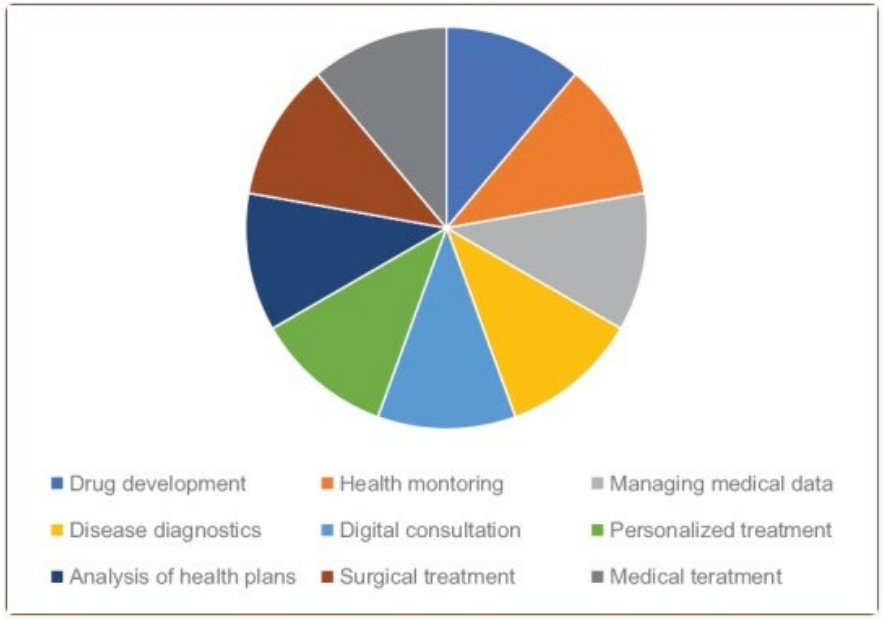
https://www.unitree.com/g1

What is driving the rapid advance in AI?

- Discussion and Demo



AI in Industry



- Healthcare – computer vision for diagnosis.
- Services – Chatbots, virtual bookings

- Energy – monitoring, predicting demand
- Financial Services – Fraud detection, Algorithmic trading
- Life Sciences - drug discovery, predicting disease spread.
- Telecommunications – network optimisation, predictive maintenance
- Agentic AI?

https://ginimachine.com/blog/fields-that-use-artificial-intelligence/

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Which AI application do you find most interesting?

A. Robotics

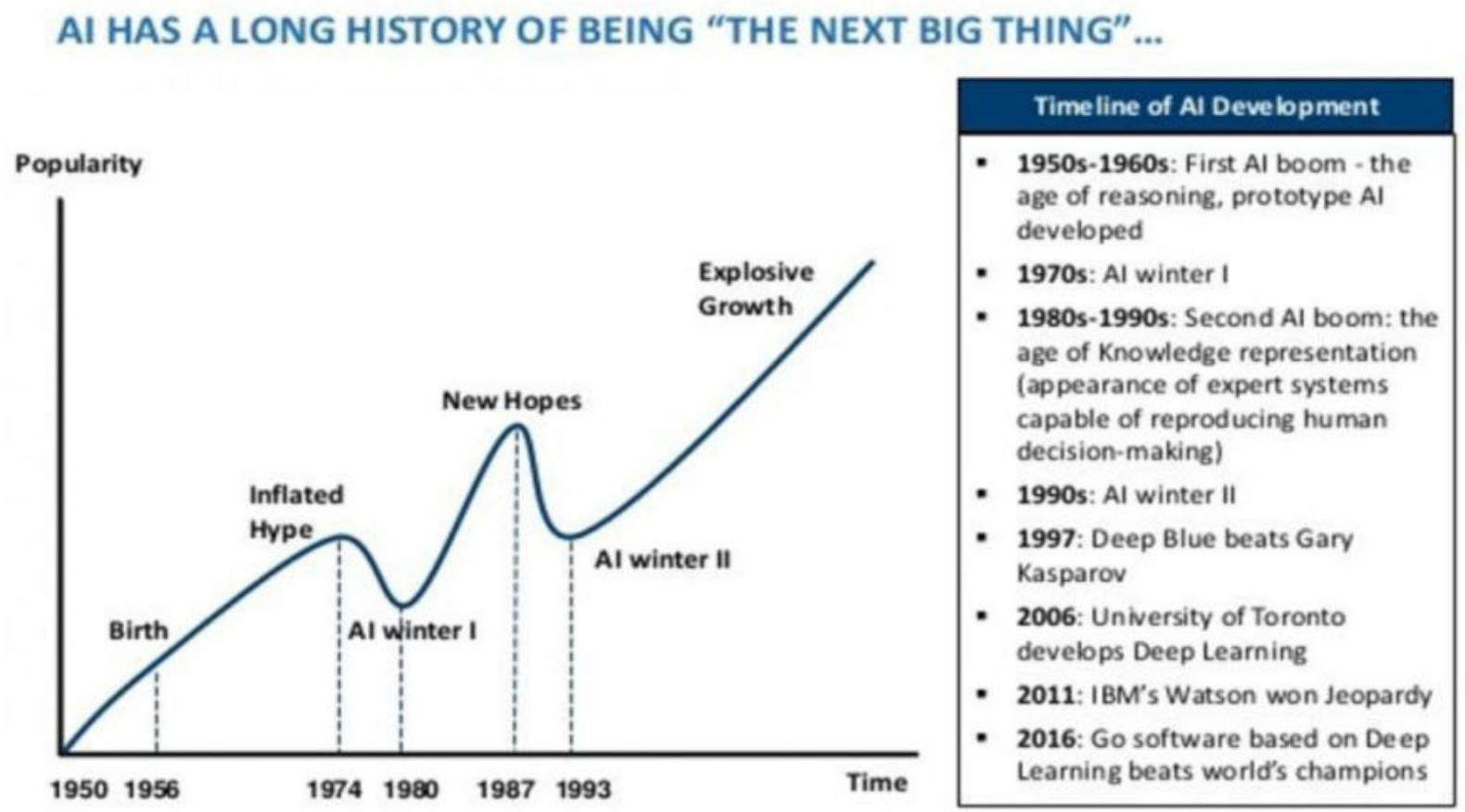
B. Computer Vision

C. Natural Language understanding

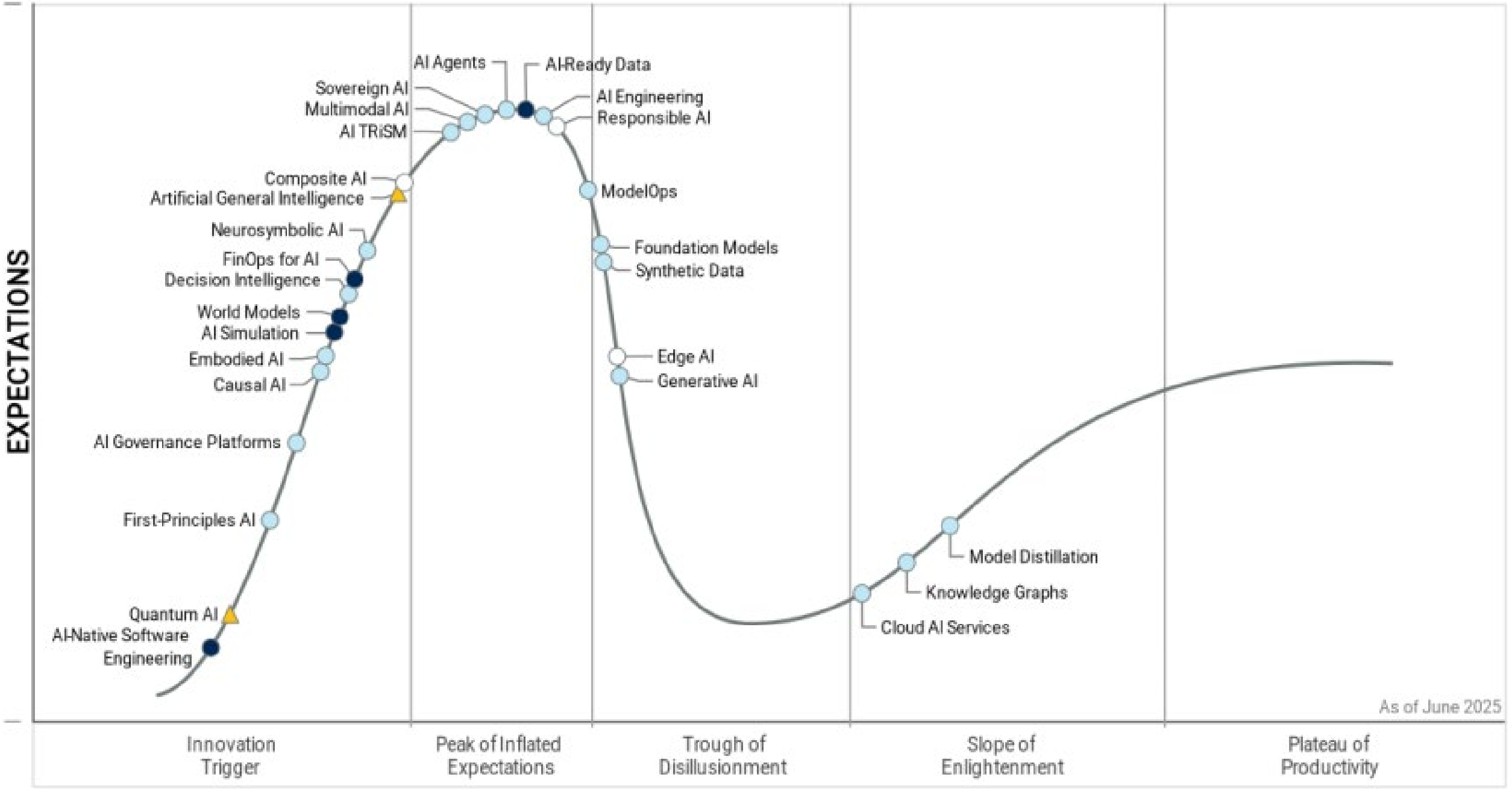
D. Gaming

E. Self Driving

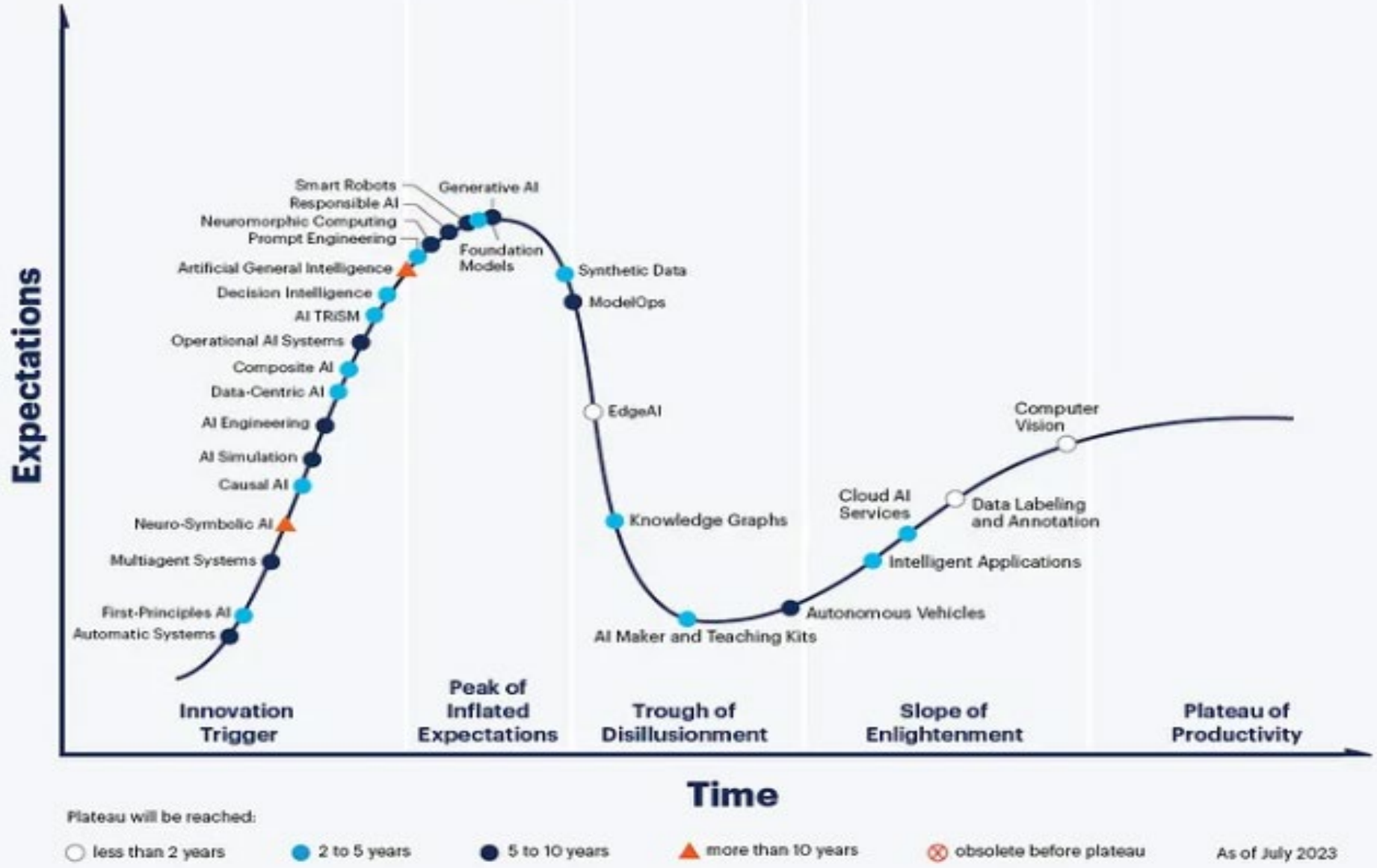
AI Promises


AI HAS A LONG HISTORY OF BEING "THE NEXT BIG THING"...
Popularity
Explosive Growth
New Hopes
Inflated Hype
AI winter II
Birth
AI winter I
1950 1956 1974 1980 1987 1993
Time
Timeline of AI Development
1950s-1960s: First AI boom - the age of reasoning, prototype AI developed
1970s: AI winter I
1980s-1990s: Second AI boom: the age of Knowledge representation (appearance of expert systems capable of reproducing human decision-making)
1990s: AI winter II
1997: Deep Blue beats Gary Kasparov
2006: University of Toronto develops Deep Learning
2011: IBM's Watson won Jeopardy
2016: Go software based on Deep Learning beats world's champions

Hype Cycle for Artificial Intelligence, 2025



Hype Cycle for Artificial Intelligence, 2023



gartner.com

Golden Age or another AI Winter?

- https://medium.com/aiguys/why-gen-ai-boom-is-fading-and-whats-next-7f1363b92696
- Lots of hype but also something is different (non toy-domain results)
- Most of the success can be attributed to advancement of computational capability and availability of ‘big data’ ie we’ve had these techniques for a while now.
- AI is always the next big thing, hence never attainable (Musk: autonomous driving next year). Should we call it Computational Intelligence to tone down the hype?
  - https://www.youtube.com/watch?v=4sCK-a33Nkk
- Money and marketing vs Scientific reality. Gartner predicts 16Trillion industry by 2030!

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Do you think we are heading towards another AI winter?

Lets discuss briefly.

Issues and concerns with AI

- Ethical/Legal implications – if AI is mainstream who is responsible?
- AI taking over jobs – including creative jobs.
- What happens if/when we reach the **singularity** – many have weighed in on this topic including – Hinton, Musk, Kurzweil etc
- Can AI achieve Consciousness? Is General AI even possible? Will it try to take over?
- Mind transfer?
- Inherent bias in data? AI favours men for jobs due to historic training data
- Can you think of others?

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What concerns you most about AI?

A. Unemployment
B. Bias
C. Privacy
D. Danger
E. Other

Advantages and benefits of AI

- Higher intelligence brings greater insight and understanding into complex problems. Did AlphaGo destroy the game of Go? Will DALL-E destroy art?
- AI/Robots can do jobs that humans cannot. Greater efficiency 24x7, productivity and hence quality of life.
- Copy/paste knowledge. Transfer of skills – humans need years.
- Aid human engineering and creativity. Bicycles for the mind.
- Autonomous wars – no/less humans involved?

Follow up

- Familiarise yourself with the book. **Artificial Intelligence: A modern approach.**
  - Read chapter 1. Introduction.
- Revise your python knowledge from first year.
- In next weeks seminar we will cover Python and work through the environment we will be using for exploring the techniques: jupyter lab, how to install libraries.

Questions Discussion?